

Scene: TX, RX, RIS with
 $N = \text{num_rows} \times \text{num_cols}$ cells

For every cell c_n : build candidate
 $\text{TX} \rightarrow c_n \rightarrow \text{RX}$ (`_ris_paths`)

both legs
unobstructed?

no

`mask[n]=False`
(obstruction)

yes

$\cos \theta_i > 0$ and
 $\cos \theta_m > 0$?

no

`mask[n]=False`
(bad angle)

yes

Compute τ_n , coef_n
(modes summed per cell)

Store $(a_n, \tau_n, \text{mask}_n)$ in the padded
`Paths` tensor (all N slots, valid or not)

`Paths.cir()` selects RIS-type slots

`cluster_ris_paths?`

no

False:
return all valid
 (a_n, τ_n) separately
 $\Rightarrow$ **genuine multipath**

yes

True (default):
coherent sum at
mean delay $\bar{\tau}$
 $\Rightarrow$ **one tap**